

Decision_Tree_Classification

```
In [6]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

```
In [7]: dataset = pd.read_csv(r'C:\Users\sande\OneDrive\Desktop\Naresh IT\data science and
```

```
In [8]: dataset
```

```
Out[8]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [9]: x = dataset.iloc[:, [2, 3]].values
y = dataset.iloc[:, -1].values
```

```
In [10]: from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size = 0.20, random_
```

Feature Scaling

```
In [11]: from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)
```

```
In [12]: from sklearn.tree import DecisionTreeClassifier
classifier = DecisionTreeClassifier(criterion='entropy', max_depth=10, random_state
classifier.fit(x_train, y_train)
```

Out[12]:

DecisionTreeClassifier		
▼ Parameters		
	criterion	'entropy'
	splitter	'best'
	max_depth	10
	min_samples_split	2
	min_samples_leaf	1
	min_weight_fraction_leaf	0.0
	max_features	None
	random_state	0
	max_leaf_nodes	None
	min_impurity_decrease	0.0
	class_weight	None
	ccp_alpha	0.0
	monotonic_cst	None

```
In [13]: from sklearn.ensemble import RandomForestClassifier
classifier = RandomForestClassifier(max_depth=4,n_estimators=30, criterion="entropy
classifier.fit(x_train, y_train)
```

Out[13]:

RandomForestClassifier		
Parameters		
	n_estimators	30
	criterion	'entropy'
	max_depth	4
	min_samples_split	2
	min_samples_leaf	1
	min_weight_fraction_leaf	0.0
	max_features	'sqrt'
	max_leaf_nodes	None
	min_impurity_decrease	0.0
	bootstrap	True
	oob_score	False
	n_jobs	None
	random_state	0
	verbose	0
	warm_start	False
	class_weight	None
	ccp_alpha	0.0
	max_samples	None
	monotonic_cst	None

```
In [14]: y_pred = classifier.predict(x_test)
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```
In [15]: from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

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[[55  3]
 [ 1 21]]
```

```
In [16]: from sklearn.metrics import accuracy_score
ac = accuracy_score(y_test, y_pred)
print(ac)
```

0.95

```
In [17]: bias = classifier.score(x_train, y_train)
bias
```

Out[17]: 0.921875

```
In [18]: variance = classifier.score(x_test, y_test)
variance
```

Out[18]: 0.95